

1. Circuits & Systems :

1-5

Network: any possible interconnection of elec. ckt elements or branches.

Circuit: is a closed energized network

All ckt's are N/w while a N/w is not necessarily a ckt.

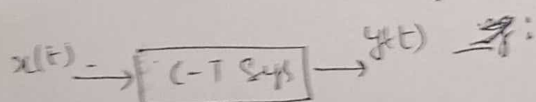
System: combi & interconnection of several components to perform a desired task.

→ System can be twoport or SISO (single I/p single O/p)

System Properties :

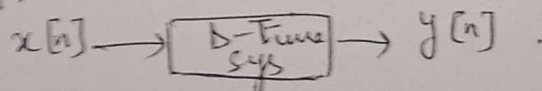
a) Continuous-time & Discrete-time Systems

Conti-time I/p gives Conti-time O/p.



- ① Series
- ② Cascade
- ③ parallel
- ④ Series-parallel
- ⑤ feed back.

In D-T Systems Dis-time S/p is transformed into D-T O/p.



b) Linear or Non-linear Sys.

A Sys is linear if it holds the principle of super-position i.e. additivity & homogeneity

otherwise Non-linear.

① $x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$

② $a x_1(t) \rightarrow a y_1(t)$

eg $y(t) = m x(t)$ (linear) (N-L) $\rightarrow y(t) = m x(t) + c$
or $y(t) = x^2(t)$

③ Time-Invariant & Time Varying Systems:

If I/P-O/P relationship doesn't change with time then Time-Invariant else time varying.
(Time shift) $x(t) \rightarrow y(t)$ } Time-Invariant
for $x(t-t_0) \rightarrow y(t-t_0)$

eg: $y(t) = \sin[x(t)]$ Time-Invariant
 $y(t) = x(2t)$ time variant.

④ Instantaneous & Dynamic Systems:

Instant $\rightarrow$ O/P is a funcⁿ of the I/P at the present time only. (memory less)

Dynamic $\rightarrow$ O/P depends on past or future values of the I/P or past O/P in addition to the present I/P.

eg: Inst $\rightarrow v(t) = R \cdot i(t)$ (Resistor)

Dyn $\rightarrow v(t) = \frac{1}{C} \int_0^t i dt + v(0^-)$ (Capacitor)

⑤ Causal & Non-Causal Systems

Causal / Non-anticipative $\rightarrow$ O/P depends only on the values of I/P at present & past.

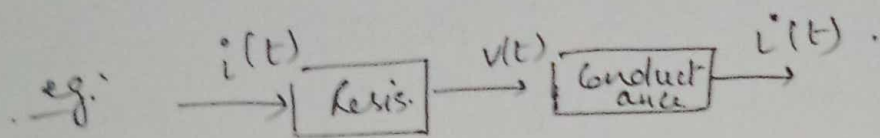
eg: $y(t) = x(t) + x(t-1)$

Non-causal $\rightarrow y(t) = x(t) + x(t+1)$.

⑥ Invertibility & Inverse Sys:

A Sys is invertible if distinct I/Ps lead to distinct O/P.

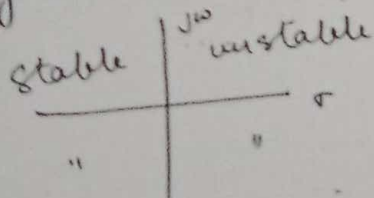
If a system is invertible, then an inverse sys² exists that when cascaded with the original sys $\rightarrow$ yields an o/p $w(t) = x(t)$.



⑦ Stability:

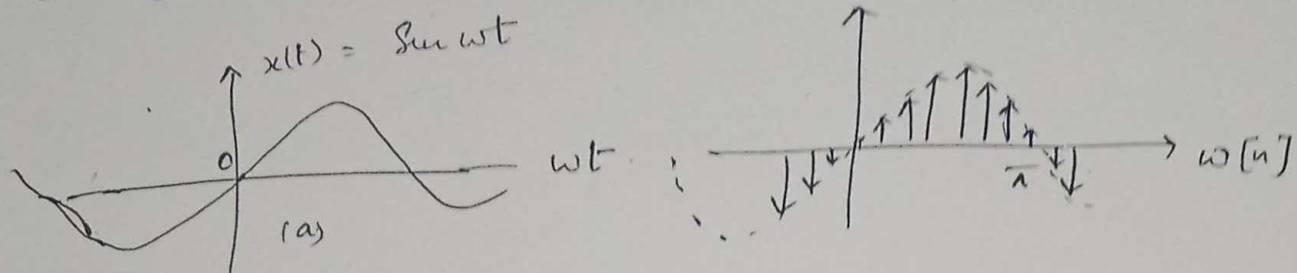
A stable sys. will remain at rest unless excited by an ext. source & will return to rest if all the excitations are removed.

* Necessary condⁿ for cont.-time sys: to be stable is that roots of char. eqⁿ have negative real parts.



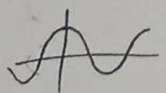
Classification of Signals : (continuous of time)
like I or voltages.

(a) Conti-Time & Discrete-Time Signals.



(b) Even & Odd signals :

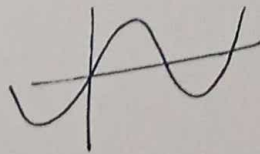
Even $\rightarrow$ if the signal is identical to its time reversed counterpart.

 $x(-t) = x(t).$

eg $x(t) = t^{2n}, \cos t.$

odd $\rightarrow x(-t) = -x(t).$

eg: $t^{2n+1}, \sin t.$ etc.



(c) Periodic & Unperiodic signals

A signal is periodic iff

$$x(t + T_0) = x(t) \quad -\infty < t < \infty$$

i.e. it repeats after certain time period.
otherwise its unperiodic.

P $\rightarrow \sin t, \cos t$

Unperio $\rightarrow e^t, t^2, t$ etc.

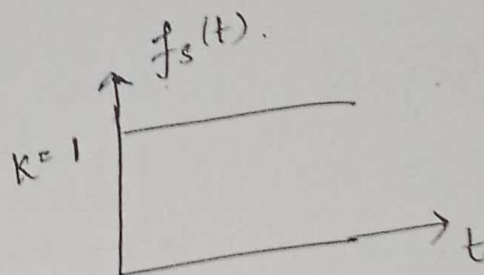
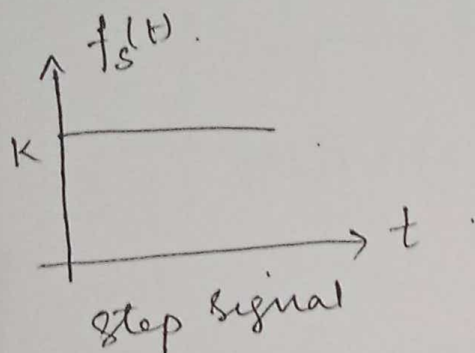
* Signal $\rightarrow$ funⁿ of time that represents a physical variable of interest associated with a system.

In elec. system excitation & response is given in terms of V & I.

Std. Signals :

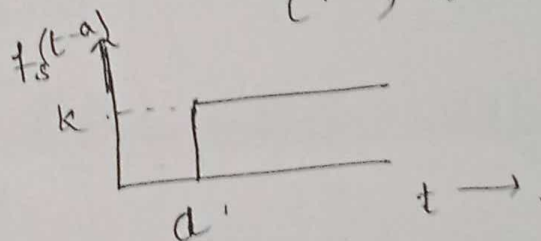
① Step Signal :

$$f_s(t) = \begin{cases} 0; & t < 0 \\ K; & t \geq 0 \end{cases}$$



$$u(t) = \begin{cases} 0; & t < 0 \\ 1; & t \geq 0 \end{cases}$$

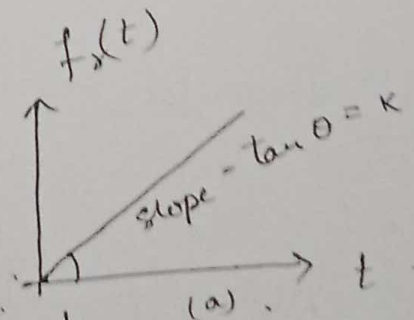
$$f_s(t-a) = \begin{cases} 0; & t < a \\ K; & t \geq a \end{cases}$$



$$* u(t-a) = \begin{cases} 0; & t < a \\ 1; & t \geq a \end{cases}$$

② Ramp Signal :

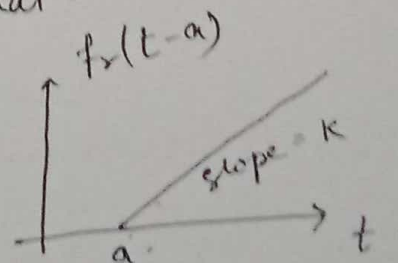
$$f_r(t) = \begin{cases} 0; & t < 0 \\ kt; & t \geq 0 \end{cases}$$



$k \rightarrow$ slope of the ramp signal.

Shifted Ramp Signal:

$$f_r(t-a) = \begin{cases} 0; & t < a \\ k(t-a); & t \geq a \end{cases}$$



$$r(t-a) = \begin{cases} 0; & t < a \\ (t-a); & t \geq a \end{cases}$$

unit Ramp Signal

$$\Rightarrow r(t) = t U(t)$$

$$r(t-a) = (t-a) U(t-a)$$

Ramp Signal

$$f_r(t) = K r(t) = K t U(t)$$

$$f_r(t-a) = K_r(t-a) = K(t-a) U(t-a)$$

③ Impulse Signal : / (Dirac Delta Signal)

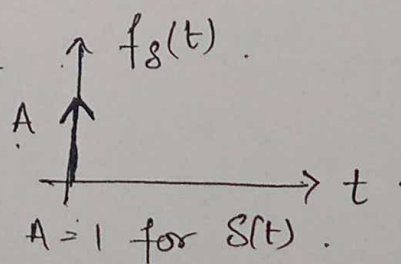
$$f_\delta(t) = \begin{cases} 0 & ; t \neq 0 \\ A & ; t = 0 \end{cases}$$

$A = \text{Area of impulse signal / strength.}$

unit Impulse Signal :

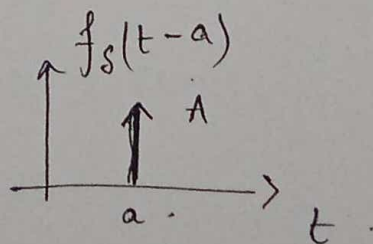
$$A = 1$$

$$\text{so } \delta(t) = \begin{cases} 0 & ; t \neq 0 \\ 1 & ; t = 0 \end{cases}$$



$$\text{Area} = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$f_\delta(t-a) = \begin{cases} 0 & ; t \neq a \\ A & ; t = a \end{cases}$$



(4a)

Relation b/w std signals :-

1. $\frac{d}{dt} (\text{step signal}) = \text{Impulse signal}$
2. $\frac{d}{dt} (\text{ramp signal}) = \text{step signal}$

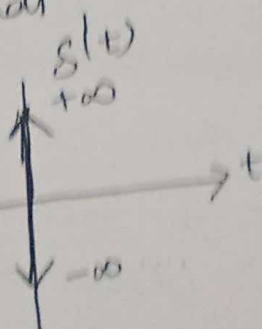
or $\int \text{Impulse} = \text{step}$
 $\int \text{step} = \text{ramp}$

Other basic signals :

a) unit doublet signal

$$s'(t) = \frac{d}{dt} [s(t)]$$

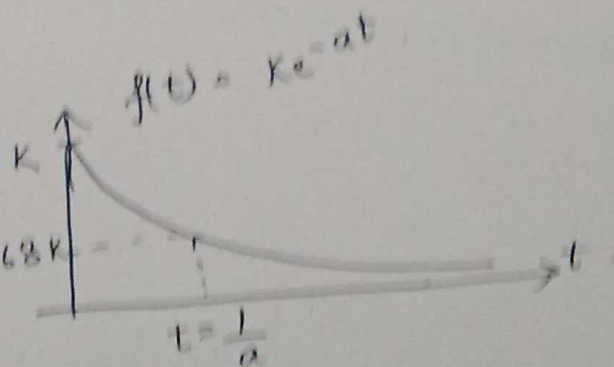
$$\begin{cases} = +\infty & t = 0 \\ = 0 & t \neq 0 \end{cases}$$



$$\begin{cases} s'(t-a) = +\infty & t = a \\ = 0 & t \neq a \end{cases}$$

b) exponential signal

$$\begin{cases} f(t) = Ke^{-at} & t \geq 0 \\ = 0 & t < 0 \end{cases}$$

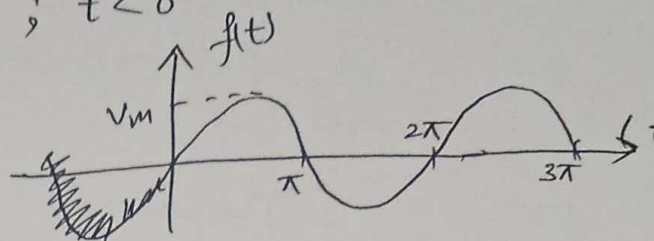


time constt = $\tau = \frac{1}{a}$

$\tau \rightarrow$ time taken to reach 63.2% of the total change from initial to final.

c Sinusoidal Signal

$$f(t) = \begin{cases} V_m \sin \omega t & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$

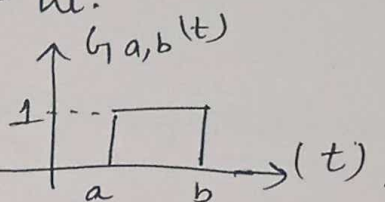


d) Gate Signal.

rectangular pulse of unit ht.

from $t=a$ to $t=b$.

$$G_{a,b}(t) = U(t-a) - U(t-b)$$



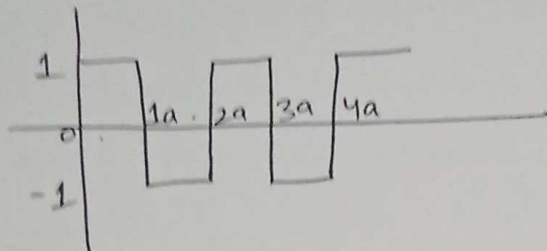
Applⁿ $\rightarrow$ any funcⁿ multiplied by a gate funcⁿ
then that funcⁿ exist only in that duration
of gate funcⁿ otherwise the signal vanishes.

~~Direct formula / K-M. formula.~~

Questions

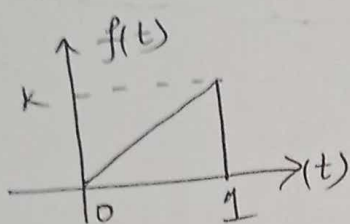
(5)

1.



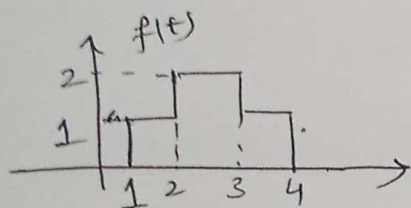
$$u(t) - 2u(t-a) + 2u(t-2a) - 2u(t-3a) + \dots$$

2.



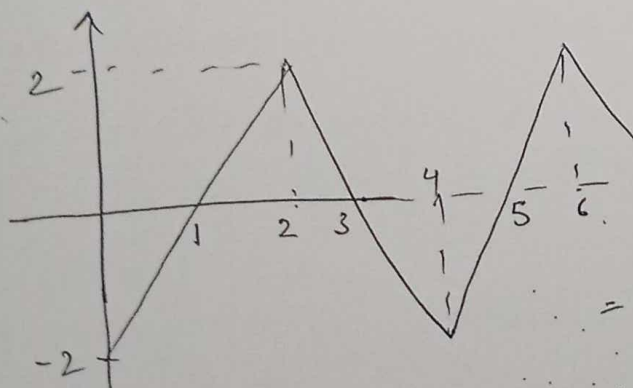
$$= K \delta(t) - K \delta(t-1) - K u(t-1) \\ = K [t u(t) - t u(t-1)]$$

3.



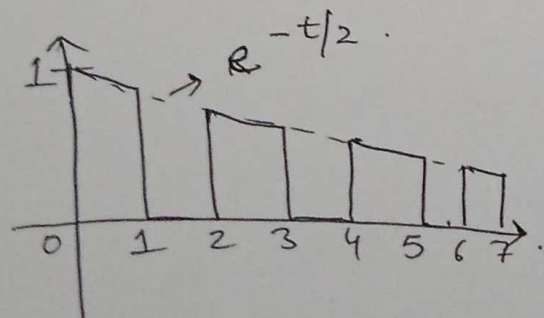
$$u(t-1) + u(t-2) - u(t-3) - u(t-4)$$

4.

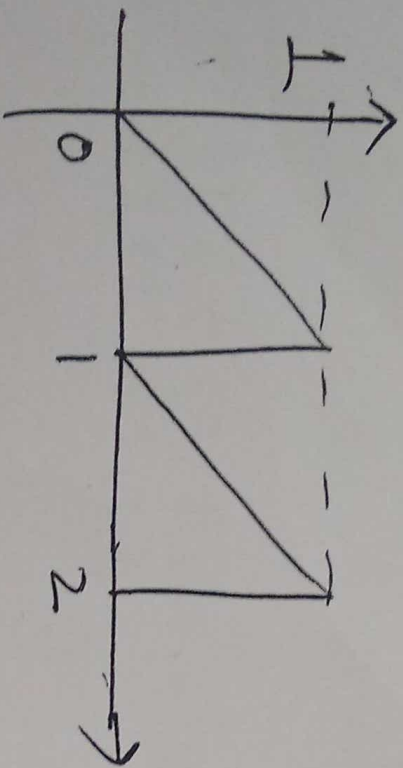


$$= 2(t-1)[u(t) - u(t-2)] \\ - 2(t-3)[u(t-2) - u(t-4)] \\ + 2(t-5)[u(t-4) - u(t-6)] \\ = 2(t-1)u(t) - 4(t-2)u(t-2) \\ + 4(t-4)u(t-4) + \dots$$

5.



$$e^{-t/2} [u(t) - u(t-1) + u(t-2) - u(t-3) + \dots]$$

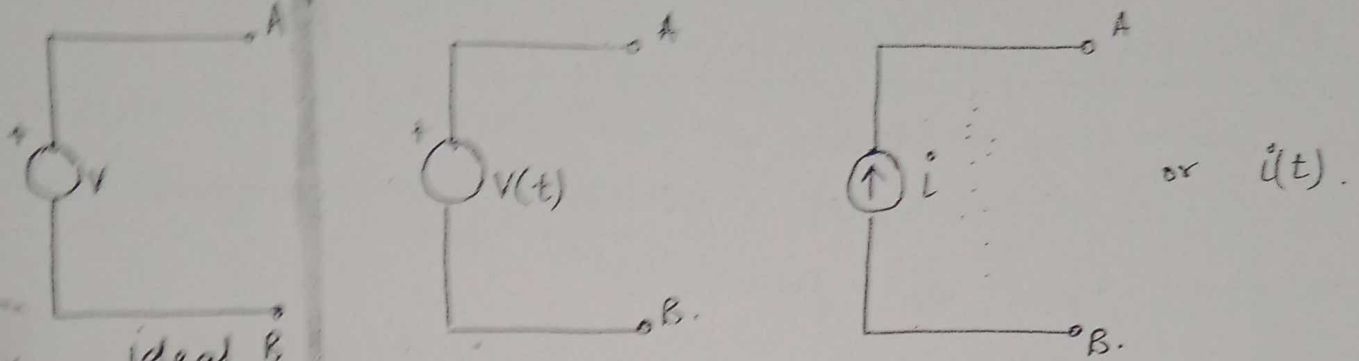


$$f(t) = x(t) - u(t-1) - x(t-2) - u(t-2)$$

7.

INDEPENDENT & DEPENDENT SOURCES:

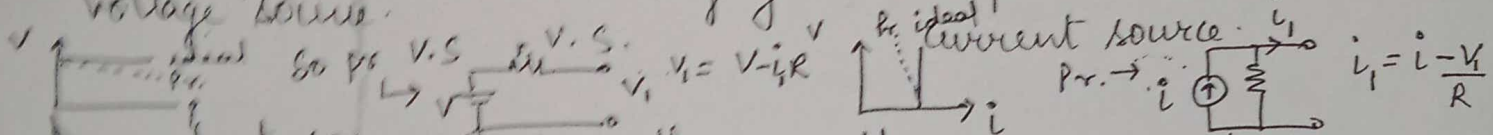
Independent sources: Those voltage or current sources, which don't depend on any other quantity in the ckt, are called independent sources.



ideal DC independent d.c. voltage source.

time varying

independent constt.



Dependent sources: are those voltage or current source which depends on some other quantity in the circuit which may be either a voltage or a current.

These sources are represented by a diamond shaped symbol.

Four possible dependent sources:-

- (i) Voltage dependent V.S.
- (ii) Current " "
- (iii) Voltage " "
- (iv) Current " "

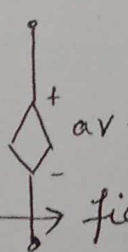
eg:- op-amp

dc shunt generator.

current source.

Common base transistor (i)

egs - ?

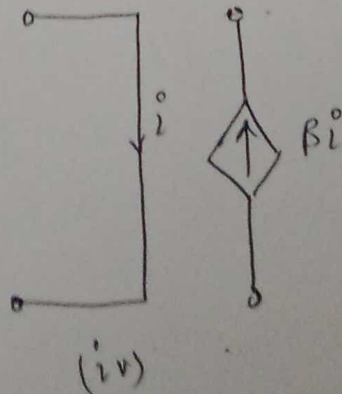
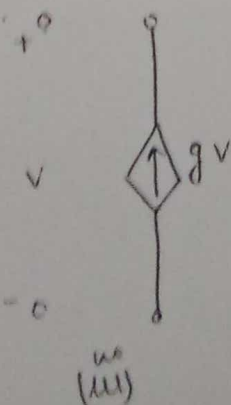
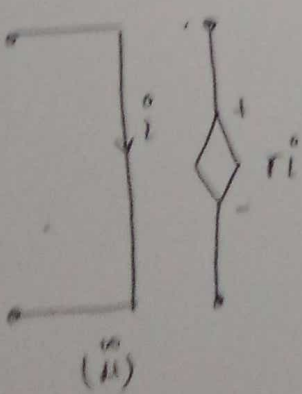


field effect transistor.

a, k

g & β →

constts of proportionality



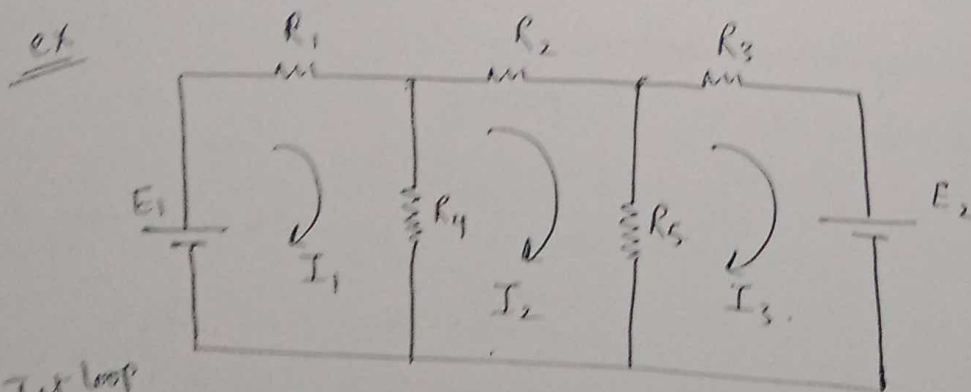
α & β have no units.

$r \rightarrow$ ohms.

$g \rightarrow$ Siemens.

Maxwell's Loop Current Method:

- $\rightarrow$ uses loop or mesh current instead of branch I.
- $\rightarrow$ I in diff. meshes is assigned continuous path so they don't split at a junction.
- $\rightarrow$ best suited when energy sources are v.s. rather than c.s.
- $\rightarrow$ Apply KVL to individual mesh.
- $b \rightarrow$ no. of independent eqⁿ when KCL & KVL is used.
- $b - j + 1 \rightarrow$ no. of eqⁿ when this method is used.
- $b \rightarrow$ no. of branch $j \rightarrow$ no. of junction.



Ist loop

$$E_1 - I_1 R_1 - (I_1 - I_2) R_4 = 0.$$

IInd loop

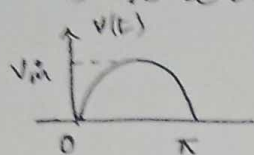
$$-(I_2 - I_1) R_4 - I_2 R_2 - (I_2 - I_3) R_5 = 0.$$

IIIrd loop

$$-R_3 I_3 - E_2 - (I_3 - I_2) R_5 = 0.$$

Questions:

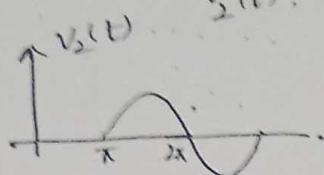
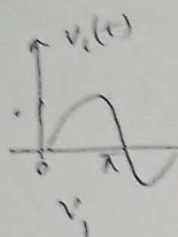
- ① Synthesize the given a single half-sine wave-form as shown in fig.



$$v(t) = v_1(t) + v_2(t)$$

$$v_1(t) = V_m \sin \omega t \cdot U(t)$$

$$v_2(t) = V_m \sin \left(\omega \left(t - \frac{T}{2} \right) \right) \cdot U \left(t - \frac{T}{2} \right)$$

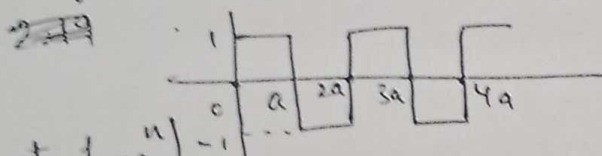


$$v(t) = V_m \left\{ \sin(\omega t) \cdot U(t) + \sin \left(\omega \left(t - \frac{T}{2} \right) \right) \cdot U \left(t - \frac{T}{2} \right) \right\}$$

using gate functⁿ

$$v(t) = V_m \sin \omega t (U(t) - U(t - T))$$

- ② Synthesize the waveform as shown in fig.



(gate funcⁿ)

$$f(t) = g_{[0,a)}(t) + g_{[a,2a)}(t) + g_{[2a,3a)}(t) + \dots$$

$$= 1 [U(t) - U(t-a)] + (-1) [U(t-a) - U(t-2a)]$$

$$+ 1 [U(t-2a) - U(t-3a)] + \dots$$

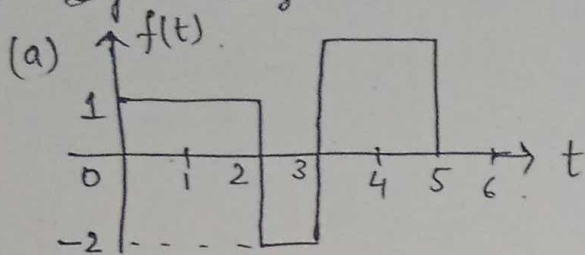
$$= U(t) - 2 U(t-a) + 2 U(t-2a) - 2 U(t-3a) + \dots$$

K.M. formula:

$$f(t) = (1-0) U(t) + (-1-1) U(t-a) + (1-(-1)) U(t-2a)$$

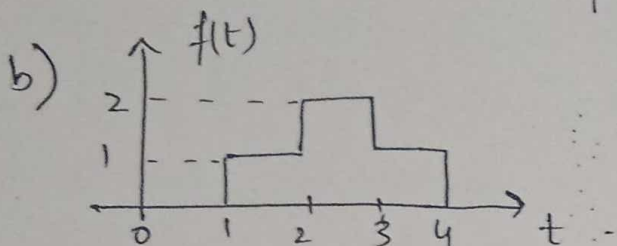
$$= 1 \cdot U(t) - 2 U(t-a) + 2 U(t-2a) + \dots$$

③ Synthesize

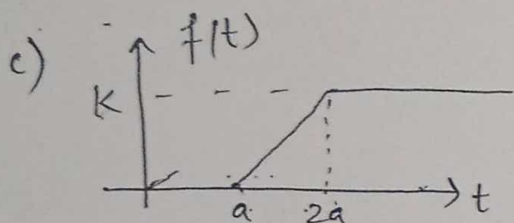


$$f(t) = u(t) - 3u(t-2) + 4u(t-3) - 2u(t-5)$$

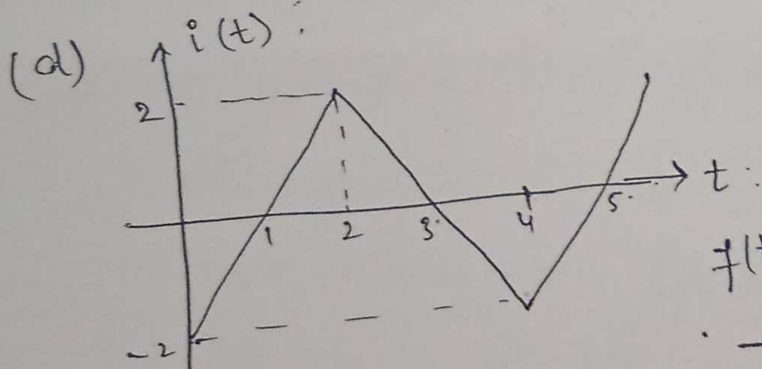
$$= 1u(t-1) + (2-1)u(t-2) + (1-2)u(t-3) + (0-1)u(t-4)$$



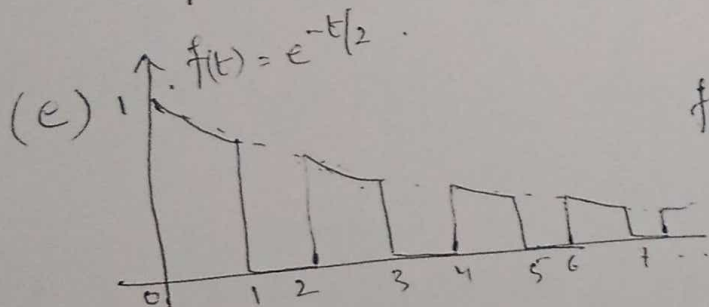
$$f(t) = u(t-1) + u(t-2) - u(t-3) - u(t-4)$$



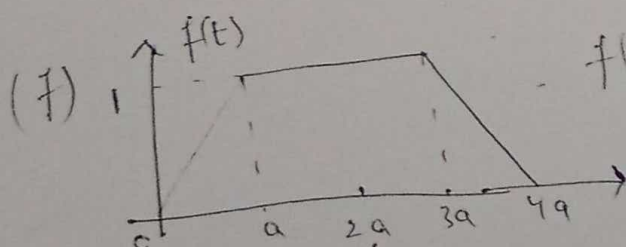
$$f(t) = \frac{K}{a}(t-a)[u(t-a) - u(t-2a)] + Ku(t-2a)$$



$$f(t) = \frac{4}{2}(t-1)[u(t) - u(t-2)] - 2(t-3)[u(t-2) - u(t-4)]$$



$$f(t) = e^{-t/2}[u(t) - u(t-1)] + e^{-t/2}[u(t-2) - u(t-3)] + \dots$$



$$f(t) = \frac{1}{a}t[u(t) - u(t-a)] + [u(t-a) - u(t-3a)] + \left(-\frac{1}{a}(t-4a)\right)[u(t-3a) - u(t-4a)]$$

$$= \frac{1}{a}tu(t) - \frac{1}{a}(t-a)u(t-a) - \frac{1}{a}(t-3a)u(t-3a) + \frac{1}{a}(t-4a)u(t-4a)$$

$$= \frac{1}{a}u(t) - \frac{1}{a}u(t-a) - \frac{1}{a}u(t-3a) + \frac{1}{a}u(t-4a)$$

Express it using ramp signals.

2. Circuit Analysis

6-17

Transient: A transition period exists whenever a ckt is switched from "old" to other.
by changing the source or in the ckt elements.
During this period branch I & element voltages change from their former to new ones.

* After transient has passed, the ckt is said to be in steady state.

linear diffⁿ eqⁿ that describes the ckt will have 2 parts to its solⁿ.

C.F. (Complimentary funcⁿ) $\rightarrow$ transient
P.I (Particular Integral) $\rightarrow$ steady state.

$\begin{matrix} R-L \\ R-L \end{matrix} \left\{ \begin{matrix} \text{I}^{\text{st}} \text{ order} \end{matrix} \right.$

$\begin{matrix} R-L-L \\ \text{Series} \\ \text{or} \\ \text{Parallel} \end{matrix} \left\{ \begin{matrix} \text{Sec order ckt.} \end{matrix} \right.$

* ckt changes are assumed to occur at time $t=0$.
& represented by a switch.

switch may close & open at $t=0$.

$t=0^-$ the instant prior to $t=0$

$t=0^+$ the instt imm. after switching.

$\begin{matrix} i_L(0^-) = i_L(0^+) \\ v_C(0^-) = v_C(0^+) \end{matrix} \left\{ \begin{matrix} \text{No change on switching} \end{matrix} \right.$

[Diff. Eqⁿ]^o of diffⁿ Eqⁿ = no. of ^{energy} storing elements.

Type I :

first order Homogeneous Diff. Eqⁿ.

$$\frac{dy(t)}{dt} + Py(t) = \underline{0} \quad P \rightarrow \text{constt.}$$

$$\frac{dy(t)}{y(t)} = -P dt$$

On integrating

$$\ln y(t) = -Pt + K' \quad \{ K' = \ln K \}$$

$$\ln y(t) = \ln e^{-Pt} + \ln K$$

$$\ln y(t) = \ln (K e^{-Pt})$$

taking Antilog :

$$\underline{y(t) = K e^{-Pt}}$$

$K \rightarrow \text{constt.}$

Type - II :

first order Non-Homo. Diff. Eqⁿ

$$\frac{dy(t)}{dt} + Py(t) = Q$$

$P \rightarrow \text{constt}$

$Q \rightarrow \text{funcⁿ of independent variable } t \text{ or a constt.}$

$$e^{Pt} \frac{dy(t)}{dt} + e^{Pt} \cdot Py(t) = Q e^{Pt}$$

$$d(x, y) = x dy + y dx$$

7

So

$$\frac{d}{dt} [y(t) \cdot e^{Pt}] = Q e^{Pt}$$

On integration:

$$y(t) \cdot e^{Pt} = \int Q e^{Pt} dt + k$$

or

$$y(t) = e^{-Pt} \int Q e^{Pt} dt + k \cdot e^{-Pt}$$

$\downarrow$ P.I. (free of k) $\downarrow$ C.F. (free of Q)

If $Q \rightarrow$ constt, then

$$y(t) = e^{-Pt} \cdot Q \cdot \frac{e^{Pt}}{P} + K e^{-Pt}$$

$$y(t) = \frac{Q}{P} + K e^{-Pt}$$

otherwise $\rightarrow$

$$y(t) = e^{-Pt} \int Q e^{Pt} dt + K e^{-Pt}$$

Type III : (Second order diff. eqn)

$$A \frac{d^2 y(t)}{dt^2} + B \frac{dy(t)}{dt} + C y(t) = 0$$

$$\text{General sol}^n : y(t) = K_1 e^{P_1 t} + K_2 e^{P_2 t}$$

K_1 & $K_2 \rightarrow$ constt.

P_1 & $P_2 \rightarrow$ roots of the quadratic eqn.

$$P_1, P_2 = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

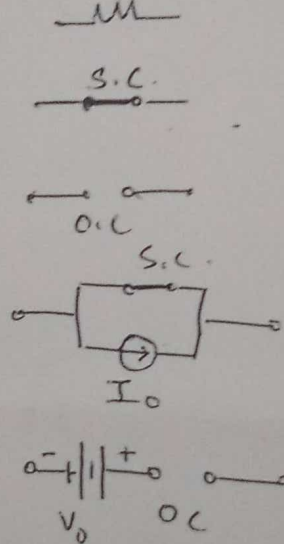
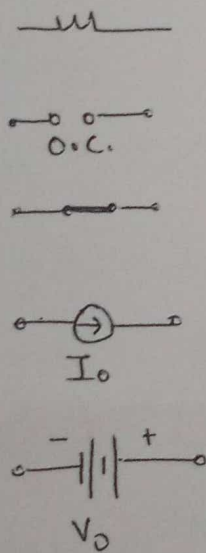
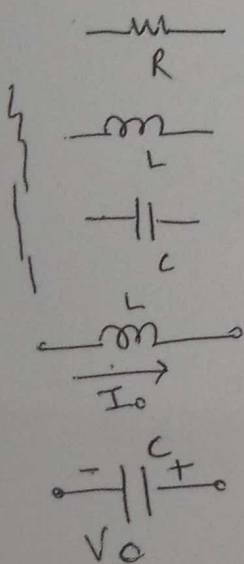
Initial Conditions in ckts:

We need initial condⁿ to evaluate the arbitrary const^s in the gen. solⁿ of Diff. "Eqⁿ".

no. of initial cond^s required = order of the diff. "Eqⁿ".

Eqⁿ. ckt⁻
at
 $t = 0^+$

Eqⁿ. ckt⁻
at
 $t = \infty$



~~TI Systems~~ {~~Linear Time Invariant~~}

Various Responses :

(a) Transient Response $\rightarrow$

values of v & i during transient period.

or part of the tot. time response that goes to zero as time becomes large.

$\rightarrow$ depends upon N/w elements alone not on forcing function (source).

$\rightarrow$ Complementary funcⁿ represent tr. response.

(b) Steady State Response $\rightarrow$

values of v & i after tr. has died or

part of tot. time response which remains after the transient has passed.

$\rightarrow$ depends on both N/w elements & forcing funcⁿ.

$\rightarrow$ P.I $\rightarrow$ represents forced or Steady State (S.S) response.

Note: So tot. response of a N/w is the sum of tr. & S.S. response.

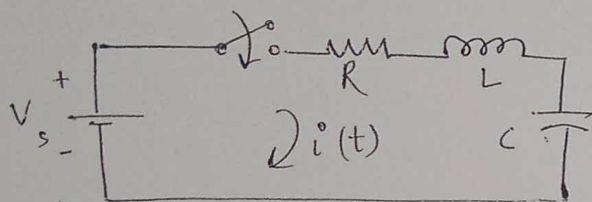
(c) Zero I/P response $\rightarrow$

values of v & i that result from initial cond^{ns} when S/P or forcing funcⁿ is zero.

(d) Zero state Response $\rightarrow$
 values of v & i for an excitation which
 is applied when all initial condⁿ $\rightarrow 0$.
 such N/w $\rightarrow$ at rest or initially relaxed.

Transient Response

(a) R-L-C series



KVL

$$L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int i dt = V_s \quad (1)$$

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{i(t)}{C} = 0$$

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{i(t)}{CL} = 0 \quad (2)$$

So $P^2 + \frac{R}{L}P + \frac{1}{CL} = 0$

$\rightarrow R = 6\Omega \quad L = 2H \quad C = 0.25F \quad V_s = 2V \quad (\text{given})$

$$P^2 + 3P + 2 = 0$$

$$P_1 = -1 \quad \& \quad P_2 = -2$$

Solⁿ of 2nd order diff. eqⁿ $\rightarrow$

$$i(t) = K_1 e^{P_1 t} + K_2 e^{P_2 t}$$

$$i(t) = K_1 e^{-t} + K_2 e^{-2t}$$

— (3)

(9)

~~initials~~Using initial condⁿ, evaluating constants k_1 & k_2

$$i(0^+) = 0 \quad (\text{switch is open at } t = 0^+)$$

4 volt across C is 0 at $t = 0^+$ as its S.C.So in eqⁿ (1), we are left with

$$\frac{di(0^+)}{dt} = \frac{V_s}{L} = \frac{2}{2} = 1 \text{ A/sec} \quad \text{--- (4)}$$

from (2)

$$\frac{d^2 i(0^+)}{dt^2} + 3 \frac{di(0^+)}{dt} + 2i(0^+) = 0$$

$$\frac{d^2 i(0^+)}{dt^2} = -3 \times 1 = -3 \text{ A/s}^2 \quad \text{--- (5)}$$

Using (4) & (5) in (3) $= 0 = k_1 + k_2$

$$\frac{di(t)}{dt} = -k_1 e^{-t} - 2k_2 e^{-2t} \Rightarrow k_1 + k_2 = 0 \quad \text{--- (a)}$$

$$\frac{di(0^+)}{dt} = -k_1 \times 1 - 2k_2 \times 1 = 1$$

$$-k_1 - 2k_2 = 1 \quad \text{--- (b)}$$

$$\text{also } \rightarrow k_1 = -k_2$$

$$+k_2 - 2k_2 = 1$$

$$-k_2 = 1$$

$$\text{So } \boxed{\begin{matrix} k_2 = -1 \\ k_1 = 1 \end{matrix}}$$

$$\text{So } i(t) = k_1 e^{-t} + k_2 e^{-2t}$$

$$\Rightarrow \boxed{i(t) = e^{-t} - e^{-2t} \text{ A}}$$

or diff.

$$\frac{di(t)}{dt} + 4i(t) = 0.$$

Solⁿ $i(t) = K e^{-4t}$

Since $V_L(t^-) = 9V$.

$$i(0^+) = \frac{1-9}{4} \text{ from (1)}$$

$$i(0^+) = -2 \text{ A.}$$

$$i(t^+) = K e^{0}.$$

$$= K \Rightarrow -2$$

So $i(t) = -2 e^{-4t} \text{ A.}$

Then $V_L(t) = 9 + \frac{1}{L} \int_0^t i(t) dt.$

$$9 + 16 \int_0^t -2 e^{-4t} dt.$$

$$V_L(t) = 9 + 16 \times (-2) \left. \frac{e^{-4t}}{-4} \right|_0^t.$$

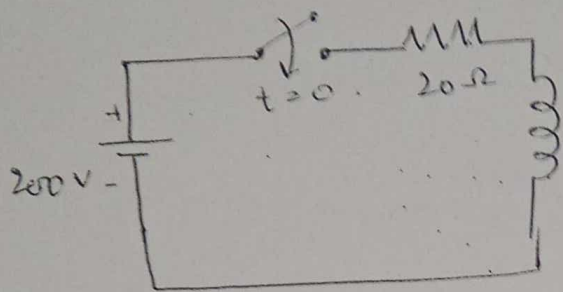
$$9 + 8 \left[e^{-4t} - 1 \right].$$

$$9 - 8 + 8e^{-4t}$$

$$\underline{\underline{V_L(t) = 1 + 8e^{-4t} \text{ V.}}}$$

(11)

Ques: A d.c. voltage of 200 V is suddenly applied to a series L-R ckt having $R = 20\ \Omega$ & inductance = 0.2 H. Determine the v.d. across the inductor at the instant of switching on + 0.02 sec. later.



$L \rightarrow 0.2\text{ H}$ at $t = 0^+$
 ~~$R \rightarrow 20\ \Omega$ at $t = 0^+$~~

so $i(0^+) = 0$.

At switching $t \rightarrow 0.02$
 so $V(0^+) = 200\text{ V}$.

Applying KVL

$$L \frac{di(t)}{dt} + Ri(t) = V$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V}{L}$$

$$i(t) = \frac{V}{R} + K e^{-\frac{R}{L}t}$$

$\frac{10}{20} = \frac{100}{200}$

$$i(0^+) = 0 = \frac{V}{R} + K$$

So $K = -\frac{V}{R}$

$$\Rightarrow i(t) = \frac{V}{R} - \frac{V}{R} e^{-\frac{R}{L}t} = 10 - 10 e^{-100t}$$

$$i(t) = \frac{V}{R} \left(1 - e^{-\frac{R}{L}t}\right) = \frac{200}{20} \left(1 - e^{-\frac{20}{0.2}t}\right) = 10 (1 - e^{-100t})\text{ A}$$

BPIT/EEE/EMEC-I

$$V_R(t) = i(t) \cdot R$$

$$= V (1 - e^{-\frac{R}{L}t}) \rightarrow V_R(0^+) = 0$$

$$= 200 (1 - e^{-100 \cdot 0.02})\text{ V}$$

$$V_R(0.02) = 200 (1 - e^{-2})$$

$$(172.9) = 8.646 \times 20\text{ V}$$

$$V_L(t) = L \cdot \frac{di(t)}{dt}$$

$$= L \cdot \frac{V}{R} \left[0 - \left(-\frac{R}{L}\right) e^{-\frac{R}{L}t} \right]$$

$$= L \cdot \frac{V}{R} \cdot \frac{R}{L} e^{-\frac{R}{L}t}$$

$$V_L(t) = V e^{-\frac{R}{L}t}$$

$$= 200 e^{-100t} \text{ V}$$

$$V_L(0.02) = 200 e^{-2} \text{ V}$$

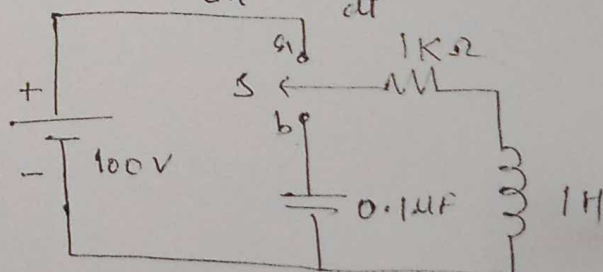
$$= \underline{\underline{27 \text{ V}}}$$

Q In the given ckt switch S is changed from position 'a' to 'b' at $t=0$. Find i , $\frac{di}{dt}$ & $\frac{d^2i}{dt^2}$ at $t=0^+$

Posⁿ a:

i behaves as s.c at $t=0$.

$$\text{So } i(0^-) = \frac{100}{1000} = 0.1 \text{ A}$$



Posⁿ b:

$$i(0^+) = i(0^-) = 0.1 \text{ A}$$

$$L \frac{di(t)}{dt} + 1000 i(t) + \frac{1}{0.1 \times 10^{-6}} \int_{-\infty}^t i(t) dt = 0 \quad \text{--- (1)}$$

Since C is uncharged so at switching it is s.c & $V_C = 0$

$$1000 i(t) + \frac{di(t)}{dt} = 0$$

$$\frac{di(0^+)}{dt} = -1000 i(0^+) = -1000 \times 0.1 = -100 \text{ A/s}$$

On differentiating (1).

$$\frac{d^2 i(t)}{dt^2} + 1000 \frac{di(t)}{dt} + \frac{1}{0.1 \times 10^{-6}} i(t) = 0$$

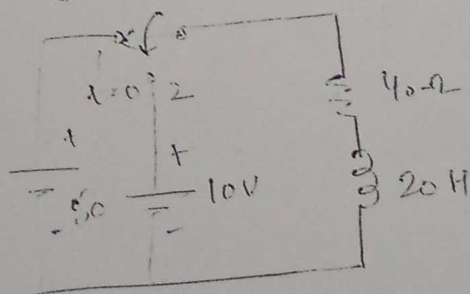
$$\frac{d^2 i(0^+)}{dt^2} = -1000 \times \frac{di(0^+)}{dt} - \frac{i(0^+)}{0.1 \times 10^{-6}}$$

$$= -1000 \times (-100) - \frac{0.1}{0.1 \times 10^{-6}}$$

$$= +10^5 - 10^6$$

$$= -9 \times 10^5 \text{ A/s}^2$$

Ques: The switch s has been in pos. 1 for a long time; it is moved to 2 at $t=0$. obtain $i(t)$; $t > 0$



$$i(0^-) = \frac{50}{40} = 1.25 \text{ A}$$

at 2

$$i(0^+) = i(0^-) = 1.25 \text{ A}$$

$$40 i(t) + \frac{20}{10^{-3}} \frac{di(t)}{dt} = \frac{10}{10^{-3}}$$

$$i(t) = \dots$$

$$\frac{di(t)}{dt} + 2i(t) = 0.5$$

$$i(t) = \frac{0.5}{2} + Ke^{-2t}$$

$$i(t) = 2.5 + Ke^{-2t}$$

$$1.25 - 2.5 = K$$

$$K = -1.25$$

Putting $i(0^+) = 1.25$ A in above eqn

$$1.25 = \frac{0.5}{2} + K$$

$$K = -1$$

$$i(t) = 0.25 + e^{-2t}$$

$$2.5 \left(1 - \frac{1}{2}e^{-\frac{5}{4}t}\right)$$

Ques Consider the ckt. & Determine

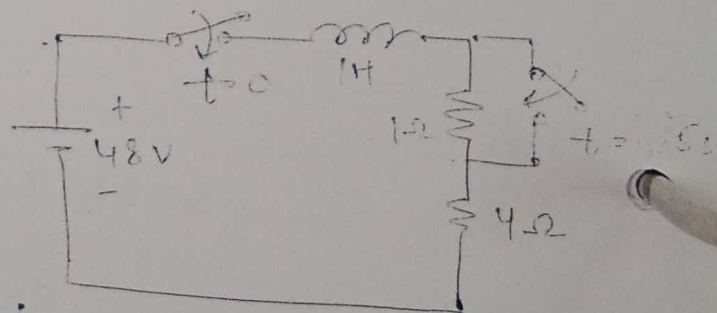
$$a) i(t), V_R(t), V_L(t)$$

$$\& \frac{di}{dt} \text{ for } t \geq 0;$$

$$b) \frac{di}{dt} \text{ at } t = 0^+;$$

$$c) \text{ the time at which } V_R = V_L;$$

d) the resistance is decreased from 5 to 4Ω at $t = 0.5$ sec. determine $i(t)$. Assume $i(0^-) = 0$.



Solⁿ

$$1. \frac{di(t)}{dt} + 5i(t) = 48$$

$$i(t) = \frac{48}{5} + Ke^{-5t}$$

$$K = -9.6$$

$$So \quad i(t) = 9.6(1 - e^{-5t}) \text{ A.}$$

$$V_L(t) = L \frac{di}{dt} = 1 \times 9.6 \times -5(-e^{-5t}) \\ = 48 e^{-5t} \text{ V.}$$

$$V_{R(t)} = Ri(t) \\ = 5 \times 9.6(1 - e^{-5t}) \\ = 48(1 - e^{-5t}) \text{ V.}$$

$$\frac{di(t)}{dt} = 48 e^{-5t} \text{ A/s.}$$

$$(b) \frac{di(0^+)}{dt} = 48 \text{ A/s.}$$

$$(c) \quad V_R(t) = V_L(t)$$

$$48(1 - e^{-5t}) = 48e^{-5t}$$

$$e^{-5t} = 1$$

$$e^{-5t} = 1$$

$$= 0$$

$$t = 0.1386 \text{ sec.}$$

$$d) \quad t = t_2 = 0.5$$

$$i(t_2) = 9.6 [1 - e^{-5 \times 0.5}] = 8.812 \text{ A.}$$

Now $R \rightarrow$ from 5Ω to 4Ω at $t = 0$.

then $i(0^-) = 8.812 \text{ A}$

$$L \frac{di(t)}{dt} + 4i(t) = 48.$$

$$i(t) = 12 + Ke^{-4t}$$

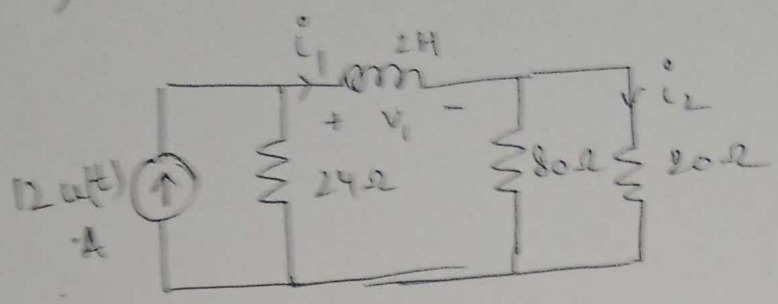
at $t = 0^-$ $i(t) = 8.812$.

so $8.812 = 12 + K$

$$K = 8.812 - 12$$
$$= -3.188$$

$$i(t) = 12 - 3.188 e^{-4t} \text{ A}$$

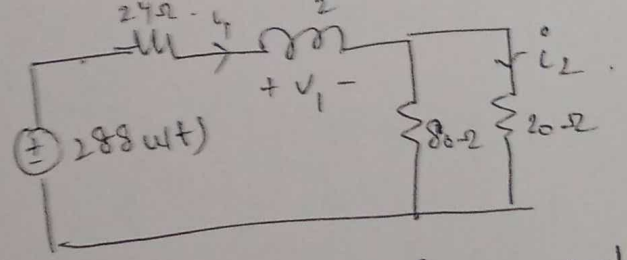
Ques For the N/w shown. find i_1 , i_2 & V_1 at
a) $t = 0^-$ b) $t = 0^+$ c) $t = \infty$ d) $t = 50 \text{ m sec}$.



Solⁿ a) at $t = 0^-$
Since $v(t) = 0$ for $t < 0$.
So $i_1(0^-) = 0$
 $i_2(0^-) = 0$
 $V_1(0^-) = 0$.

b) at $t = 0^+$
~~C.S~~ $\rightarrow$ ~~12 u(t)~~ Volt. Source.

$V.S = 12 u(t) \times 24 = 288 u(t)$



$$288 u(t) = 24 i_1 + 2 \frac{di_1}{dt} + 80(i_1 - i_2)$$

$$i_2 = \frac{80}{20+80} \times i_1$$

$$i_2 = \frac{80 i_1}{100} = 0.8 i_1$$

$$288 u(t) = 24 \dot{i}_1 + 2 \frac{d\dot{i}_1}{dt} + 80(\dot{i}_1 - .8 \dot{i}_1)$$

$$= \frac{2 d\dot{i}_1}{dt} + 24 \dot{i}_1 + 16 \dot{i}_1$$

$$288 u(t) = 2 \frac{d\dot{i}_1}{dt} + 40 \dot{i}_1$$

$$\frac{d\dot{i}_1}{dt} + 20 \dot{i}_1 = 144 u(t)$$

$$\dot{i}_1(t) = \frac{144}{20} + K e^{-20t}$$

$$t = 0^+ \quad L \rightarrow 0 \cdot L$$

$$\text{So } \dot{i}_1(0^+) = 0$$

$$K = -\frac{144}{20}$$

$$\text{So } \dot{i}_1(t) = 7.2 (1 - e^{-20t})$$

$$\dot{i}_2(t) = .8 \dot{i}_1 = .8 \times 7.2 (1 - e^{-20t})$$

$$= 5.76 (1 - e^{-20t})$$

$$\dot{i}_1(0^+) = 0$$

$$\dot{i}_2(0^+) = 0$$

$$V_1(0^+) = 288$$

$$\text{at } t = \infty$$

$$\dot{i}_1(\infty) = 7.2 \text{ A}$$

$$\dot{i}_2(\infty) = 5.76 \text{ A}$$

$$V_1(\infty) = L \frac{d\dot{i}_1(\infty)}{dt} = 0$$

$$\text{as } L \Rightarrow \underline{\underline{S.C}}$$

at $t = 50 \text{ msec}$

$$i_1 = 7.2 (1 - e^{-20 \times 50 \times 10^{-3}})$$

$$= 7.2 (1 - e^{-1}) = 4.55 \text{ A}$$

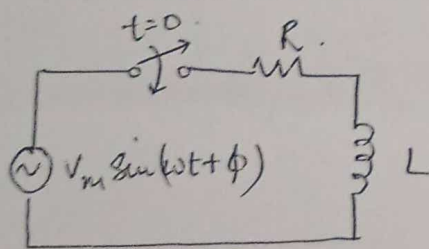
$$i_2 = 5.76 (1 - e^{-1}) = 3.64 \text{ A}$$

$$V_1 = L \left. \frac{di_1}{dt} \right|_{t=50 \text{ ms}} = 2 \times 7.2 \times 20 e^{-20 \times 50 \times 10^{-3}}$$

$$= 105.95 \text{ V}$$

Ques

R-L Series with sinusoidal excitation:



$$Ri + L \frac{di}{dt} = V_m \sin(\omega t + \phi)$$

$$\frac{di}{dt} + \frac{R}{L} i = \frac{V_m}{L} \sin(\omega t + \phi)$$

$$i = k e^{-\frac{R}{L} t} + e^{-\frac{R}{L} t} \int \frac{V_m}{L} \sin(\omega t + \phi) e^{\frac{R}{L} t} dt$$

$$i' = e^{-\frac{R}{L} t} \int \frac{V_m}{L} \sin(\omega t + \phi) e^{\frac{R}{L} t} dt$$

$$= e^{-\frac{R}{L} t} \frac{V_m}{L} \left[\frac{e^{\frac{R}{L} t}}{\left(\frac{R}{L}\right)^2 + \omega^2} \left(\frac{R}{L} \sin(\omega t + \phi) - \omega \cos(\omega t + \phi) \right) \right]$$

$$= \frac{V_m}{L \sqrt{\left(\frac{R}{L}\right)^2 + \omega^2}} \times \frac{R}{L} \sin(\omega t + \phi) - \frac{V_m}{L \sqrt{\left(\frac{R}{L}\right)^2 + \omega^2}} \times \omega \cos(\omega t + \phi)$$

$$= \frac{V_m}{R^2 + \omega^2 L^2} \left[R \sin(\omega t + \phi) - \omega L \cos(\omega t + \phi) \right]$$

$$\text{Let } R = C \cos \theta = \omega L = C \sin \theta$$

$$\text{So } \frac{V_m C}{R^2 + \omega^2 L^2} \left[\sin(\omega t + \phi) \cos \theta + \sin \theta \cos(\omega t + \phi) \right]$$

$$\checkmark \quad C = \sqrt{R^2 + \omega^2 L^2} \quad \theta = -\tan^{-1} \frac{\omega L}{R}$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin(\omega t + \phi + \theta) \right]$$

$$\frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin \omega t + \phi - \tan^{-1} \frac{\omega L}{R} \right]$$

$$\text{So total } i = K e^{-\frac{R}{L} t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin \left(\omega t + \phi - \tan^{-1} \frac{\omega L}{R} \right) \right]$$

$$\text{at } t = 0 \quad i(0^+) = 0$$

$$\text{So } i(0^+) = 0 = K + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin \left(\phi - \tan^{-1} \frac{\omega L}{R} \right) \right]$$

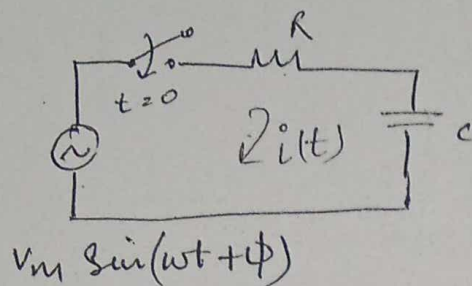
$$K = - \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin \left(\phi - \tan^{-1} \frac{\omega L}{R} \right) \right]$$

$$i(t) = \frac{V_m}{Z} \left[\sin(\omega t + \phi + \theta) - \sin(\phi + \theta) e^{-\frac{R}{L} t} \right]$$

$$Z = \sqrt{R^2 + (\omega L)^2}$$

$$\theta = -\tan^{-1} \frac{\omega L}{R}$$

R-C ckt having sinusoidal excitation:



on Diff.

$$R i(t) + \frac{1}{C} \int i dt = v_m \sin(\omega t + \phi)$$

$$\frac{di(t)}{dt} + \frac{1}{RC} \int i dt = \frac{v_m}{R} \sin(\omega t + \phi)$$

$$i(t) = i_c(t) + i_p(t)$$

$$= K e^{-\frac{t}{RC}} + e^{-\frac{t}{RC}} \int \frac{v_m}{R} \sin(\omega t + \phi) \cdot e^{\frac{t}{RC}} dt$$

$$i_p(t) = \frac{v_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left(\omega t + \phi + \tan^{-1} \frac{1}{\omega CR}\right)$$

Since $C \rightarrow \text{S.C. at } t=0$

So $i(0^+) = \frac{v_m \sin \phi}{R}$

$$\frac{v_m \sin \phi}{R} = K + \frac{v_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left(\phi + \tan^{-1} \frac{1}{\omega CR}\right)$$

$$K = \frac{v_m \sin \phi}{R} - \frac{v_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left(\phi + \tan^{-1} \frac{1}{\omega CR}\right)$$

So $i(t) = \left[\frac{v_m \sin \phi}{R} - \frac{v_m}{Z} \sin(\phi + \theta) \right] e^{-\frac{t}{RC}} + \frac{v_m}{Z} \sin(\omega t + \phi + \theta)$

Q A volt $v = 300 \sin 314t$ is applied at $t = 2.14 \text{ msec}$ to a series R-C ckt having resistance 10Ω & $C = 200 \mu\text{F}$. Find an expression for I . also $i(t)$ for $t = 1 \text{ msec}$.

Solⁿ

$$t = 2.14 \text{ msec}$$

$$\phi = \omega t = 2.14 \times 10^{-3} \times 314$$

$$\phi = 0.672 \text{ rad.}$$

$$Z = \sqrt{10^2 + \left(\frac{1}{314 \times 200 \times 10^{-6}}\right)^2} = 18.8 \Omega$$

$$i_c(t) = K e^{-\frac{t}{RC}} = K e^{-\frac{t}{10 \times 200 \times 10^{-6}}}$$

$$i_c(t) = K e^{-500t}$$

$$\begin{aligned} i_p(t) &= \frac{V_m}{Z} \sin(\omega t + \phi + \tan^{-1} \frac{1}{\omega CR}) \\ &= \frac{300}{18.8} \sin[314t + 0.672 + \tan^{-1} \left(\frac{1}{314 \times 200 \times 10^{-6}} \times 10 \right)] \\ &= 15.96 \sin(314t + 0.672 + 1.59) \end{aligned}$$

$$\therefore i(t) = K e^{-500t} + 15.96 \sin(314t + 0.672 + 1.59)$$

Capacitor behaves as S.C at switching.

$$i(2.14 \text{ msec}) = \frac{300 \sin(314 \times 2.14 \times 10^{-3})}{10} = 18.67 \text{ A}$$

$$\begin{aligned} \therefore 18.67 &= K(1) + 15.96 \sin(0.672 + 1.59) \\ K &= 6.38 \end{aligned}$$

$$\therefore i(t) = 6.38 e^{-500t} + 15.96 \sin(314t + 2.262)$$

After 1 msec $\rightarrow$

$$\begin{aligned} i &= 6.38 e^{-500 \times 10^{-3}} + 15.96 \sin(314 \times 10^{-3} + 2.262) \\ &= 3.87 + 8.55 \\ &= \underline{\underline{12.42 \text{ A}}} \end{aligned}$$

Solⁿ

$$i_1(t) = e^{-pt} \int Q(t) e^{pt} dt + K e^{-pt}$$

$$= e^{-3t} \int \frac{-e^{-t}}{10} e^{3t} dt + K e^{-3t}$$

$$i_1(t) = \frac{-e^{-3t}}{10} \int e^{2t} dt + K e^{-3t}$$

$$= \frac{e^{-3t}}{10} \times \frac{e^{2t}}{2} + K e^{-3t}$$

$$i_1(t) = \frac{e^{-t}}{20} + K e^{-3t} \quad \text{--- (5)}$$

Since at $t = 0^+$ $C \rightarrow s.c.$

$$\text{So } i_1(0^+) = i(0^+)$$

$$i(0^+) = \frac{V_1(0^+)}{R_1} = \frac{e^{-0}}{10} = \frac{1}{10} A \quad \text{--- (6)}$$

Put (6) in (5)

$$\frac{1}{10} = \frac{-1}{20} + K \quad K = \frac{3}{20}$$

$$i_1(t) = \frac{-e^{-t}}{20} + \frac{3}{20} e^{-3t}$$

$$V_2(t) = \frac{1}{C} \int i_1(t) dt$$

$$= \frac{1}{20} \int \left(\frac{-1}{20} e^{-t} + \frac{3}{20} e^{-3t} \right) dt = e^{-t} - e^{-3t} V$$

$$V_2(0^+) = 0$$

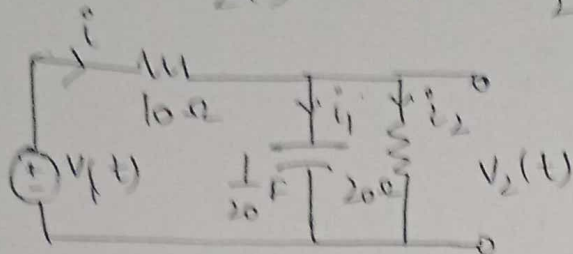
$$\frac{dV_2(0^+)}{dt} = -e^{-t} + 3e^{-3t} \Big|_{t=0} = 2V/s$$

$$\frac{d^2 V_2(0^+)}{dt^2} = -8V/s^2$$

$$\frac{d^3 V_2(0^+)}{dt^3} = 26V/s^3$$

Ques In the given ckt $V_1(t) = e^{-t}$ for $t \geq 0$

Find $V_2(t)$ & $V_2, \frac{dV_2}{dt}, \frac{d^2V_2}{dt^2}$ & $\frac{d^3V_2}{dt^3}$ at $t = 0^+$.



Solⁿ $i(t) = i_1(t) + i_2(t) \quad \text{--- (1)}$

$V_1(t) = 10 i(t) + V_2(t) \quad \text{--- (2)}$

$V_2(t) = 20 i_2(t) = \frac{1}{C} \int_0^t i_1(t) dt \quad \text{--- (3)}$

$e^{-t} = 10 (i_1(t) + i_2(t)) + V_2(t)$

$V_1(t) = 10 (i_1(t) + i_2(t)) + 20 i_2(t)$

$= 10 (i_1(t)) + (10 + 20) i_2(t)$

$= 10 i_1(t) + 30 \left(\frac{1}{R_2 C} \int i_1(t) dt \right)$

$10 i_1(t) + 30 \times \frac{1}{20 \times \frac{1}{20}} \int i_1(t) dt$

$e^{-t} = 10 i_1(t) + 30 \int i_1(t) dt \quad \text{--- (4)}$

On Differentiating (4)

$\frac{d}{dt} e^{-t} = 10 \frac{di_1(t)}{dt} + 3 i_1(t) = -\frac{e^{-t}}{10}$

S. Laplace Transform :

Characteristics :

Signal $\rightarrow$ (1) - (4)
Transient (1 - 6)

- ① gives total solⁿ in 1 operation.
- ② Initial conditions are automatically specified in the transform eqⁿ in the first step itself.
- ③ It transforms integro-differential eqⁿs in time domain to simple algebraic eqⁿs in s or frequency domain.
- ④ Operation is simple as logarithmic operation.

Def :

for the time funcⁿ $f(t)$ which is 0 for $t < 0$ &

Satisfy $\int_0^{\infty} |f(t) e^{-\sigma t}| dt < \infty$

for real & positive σ

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t) e^{-st} dt$$

s = Lap. operator

$s = \sigma + j\omega$ (Complex)

Inverse Laplace Transformation

$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} F(s) e^{st} ds$$

Properties:

a) Linearity

$$\textcircled{1} \quad \mathcal{L}[k f(t)] = k F(s)$$
$$\textcircled{2} \quad \mathcal{L}[f_1(t) \pm f_2(t)] = F_1(s) \pm F_2(s).$$

b) Time differentiation:

$$\mathcal{L}\left[\frac{d f(t)}{dt}\right] = s F(s) - \lim_{t \rightarrow 0} f(t) = s F(s) - f(0^+)$$

Proof

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t) e^{-st} dt.$$

Let $f(t) = u$; then $\left[\frac{d f(t)}{dt}\right] \cdot dt = du$.

& $e^{-st} \cdot dt = dv$; then $v = -\frac{1}{s} e^{-st}$.

So $F(s) = \int_0^{\infty} v dv = v \cdot v \Big|_0^{\infty} - \int_0^{\infty} v du$.

$$= f(t) \cdot \left(-\frac{1}{s} e^{-st}\right) \Big|_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st}\right) \left[\frac{d f(t)}{dt}\right] dt$$

$$= \frac{1}{s} \left[-f(\infty) e^{-\infty} + f(0^+) e^{-0} \right]$$

$$+ \frac{1}{s} \int_0^{\infty} e^{-st} \left[\frac{d f(t)}{dt}\right] dt$$

$$F(s) = \frac{1}{s} f(0^+) + \frac{1}{s} \mathcal{L}\left[\frac{d f(t)}{dt}\right].$$

So $\mathcal{L}\left[\frac{d f(t)}{dt}\right] = s F(s) - f(0^+) \leftarrow$

$$\int u dv = uv - \int v du \quad (2)$$

$$\begin{aligned} \mathcal{L} \left[\frac{d^2 f(t)}{dt^2} \right] &= \mathcal{L} \left[\frac{d}{dt} \left\{ \frac{d f(t)}{dt} \right\} \right] \\ &= s \mathcal{L} \left[\frac{d f(t)}{dt} \right] - \left. \frac{d f(t)}{dt} \right|_{t=0} \\ &= s [s F(s) - f(0^+)] - f'(0^+) \\ &\Rightarrow s^2 F(s) - s f(0^+) - f'(0^+) \end{aligned}$$

c) Time-Integration:

$$\mathcal{L} \left[\int_0^t f(t) dt \right] = \frac{F(s)}{s}$$

Proof $\mathcal{L} \left[\int_0^t f(t) dt \right] = \int_0^\infty \left[\int_0^t f(t) dt \right] e^{-st} dt$

So $\int_0^\infty u dv = u v \Big|_0^\infty - \int_0^\infty v du$

$$= \left[\int_0^t f(t) dt \right] \left(-\frac{1}{s} \right) e^{-st} \Big|_0^\infty$$

$$= \int_0^\infty \left(-\frac{1}{s} \right) e^{-st} f(t) dt$$

$$= \frac{F(s)}{s}$$

for $\mathcal{L} \left[\int f(t) dt \right] = \frac{F(s)}{s} - \frac{f'(0)}{s}$

d) $\mathcal{L}[t \cdot f(t)] = - \frac{d F(s)}{ds}$ freq - Differentiation

Proof

$$\frac{d}{ds} F(s) = \frac{d}{ds} \int_0^{\infty} f(t) e^{-st} dt$$

$$= \int_0^{\infty} f(t) \left(\frac{d}{ds} e^{-st} \right) dt$$

$$= \int_0^{\infty} f(t) e^{-st} (-t) dt = - \int_0^{\infty} t \cdot f(t) e^{-st} dt$$

$$= - \mathcal{L}[t \cdot f(t)]$$

e) freq - Integration:

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_0^{\infty} F(s) \cdot ds$$

(f) Shifting Theorem:

$$\mathcal{L}[f(t-a) \cdot u(t-a)] = e^{-as} \cdot F(s) \quad (\text{time})$$

$$\mathcal{L}[e^{-at} f(t)] = \int_0^{\infty} e^{-at} f(t) e^{-st} dt = F(s+a)$$

g) Initial value Theorem : (i) Theorem for Periodic funcⁿ (3)
 L.T. of a periodic funcⁿ with Time period T

$$f(0^+) = \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} \left[s \cdot F(s) \right] = \frac{1}{1 - e^{-Ts}} \text{ d[first cycle of funcⁿ].}$$

h) Final value Theorem : (3)

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} \left[s \cdot F(s) \right]$$

L.T. of Common funcⁿ :

1) $\mathcal{L}[e^{at}] = \frac{1}{s-a}$

$$\begin{aligned} \int_0^{\infty} e^{at} e^{-st} dt &= \int_0^{\infty} e^{-(s-a)t} dt \\ &= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} \end{aligned}$$

$$= 0 + \frac{1}{s-a} = \underline{\underline{\frac{1}{s-a}}}$$

2) $\mathcal{L}[u(t)] = \frac{1}{s}$

$u(t) = 1 \quad t > 0$

$$\int_0^{\infty} e^{-st} dt = \frac{1}{s}$$

$$3) \mathcal{L}[\sin \omega t] =$$

$$\int_0^{\infty} e^{-st} \sin \omega t dt = \left[\frac{e^{-st}}{s^2 + \omega^2} \left(-s(\sin \omega t) - \omega \cos \omega t \right) \right]_0^{\infty}$$

$$= - \left[\frac{1}{s^2 + \omega^2} [-\omega] \right]$$

$$= \frac{\omega}{s^2 + \omega^2}$$

$$4) \mathcal{L}[\cos \omega t]$$

$$\int_0^{\infty} e^{-st} \cos \omega t dt = \left[\frac{e^{-st}}{s^2 + \omega^2} \left(-s \cos \omega t + \omega \sin \omega t \right) \right]_0^{\infty}$$

$$= 0 - \left[\frac{1}{s^2 + \omega^2} (-s) \right]$$

$$= \frac{s}{s^2 + \omega^2}$$

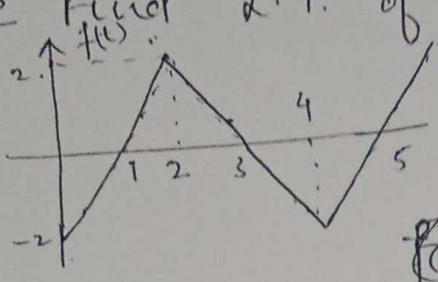
$$5) \mathcal{L}[t^n] = \frac{n!}{s^{n+1}} \quad \checkmark$$

$$\int_0^{\infty} t e^{-st} dt \quad \mathcal{L}[t] = \frac{1}{s^2}$$

Ques Find L.T. of foll:

(4)

(a)



$$f'(t) = (2t-2)[u(t) - u(t-2)] + (-2t+6)[u(t-2) - u(t-4)]$$

$$= 2tu(t) - 2u(t) - 4(t-2)u(t-2) + 2(t-4)u(t-4) + 2u(t-4)$$

$$F'(s) = \frac{2}{s^2} - \frac{2}{s} - \frac{4e^{-2s}}{s^2} + \frac{2e^{-4s}}{s^2} + \frac{2e^{-4s}}{s}$$

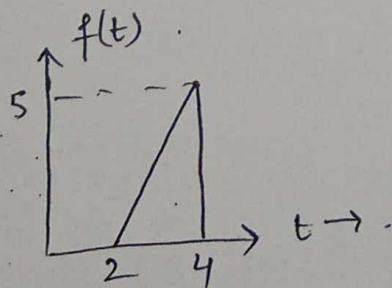
$$= \frac{2}{s^2} [1 - 2e^{-2s} + e^{-4s}] - \frac{2}{s} [1 - e^{-4s}]$$

$$F'(s) = \frac{2}{s^2} (1 - e^{-2s})^2 - \frac{2}{s} (1 - e^{-4s})$$

$$F(s) = \frac{1}{1 - e^{-4s}} \cdot F'(s) = \frac{2(1 - e^{-2s})^2}{s^2(1 - e^{-4s})} - \frac{2(1 - e^{-4s})}{s(1 - e^{-4s})}$$

$$= \frac{2}{s^2} \left(\frac{(1 - e^{-2s})^2}{1 - e^{-4s}} - \frac{1}{s} \right) \checkmark$$

Ques



$$f(t) = \frac{5}{2}(t-2)[u(t-2) - u(t-4)]$$

$$= \frac{5}{2}(t-2)u(t-2) - \frac{5}{2}(t-2)u(t-4)$$

$$= \frac{5}{2}(t-2)u(t-2) - \frac{5}{2}(t-4+2)u(t-4)$$

$$= \frac{5}{2}(t-2)u(t-2) - \frac{5}{2}(t-4)u(t-4) - 5u(t-4)$$

$$\mathcal{L}\{f(t)\} = \frac{5}{2} \frac{e^{-2s}}{s^2} - \frac{5}{2} \frac{e^{-4s}}{s^2} - \frac{5e^{-4s}}{s}$$

$$= \frac{5e^{-2s}}{2s^2} [1 - e^{-2s} - 2se^{-2s}] \quad (4)$$

Ques Without finding L^{-1} , find $f(0^+)$ & $f(\infty)$.

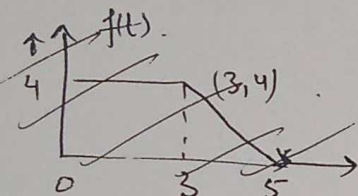
a) $F(s) = \frac{5s^3 - 1600}{s(s^3 + 18s^2 + 9s + 800)}$

$s \cdot F(s) = \lim_{s \rightarrow \infty} \frac{s^3}{s^3} \left[\frac{5 - \frac{1600}{s^3}}{1 + \frac{18}{s} + \frac{9}{s^2} + \frac{800}{s^3}} \right]$

$$\lim_{s \rightarrow \infty} \frac{5}{1} = 5$$

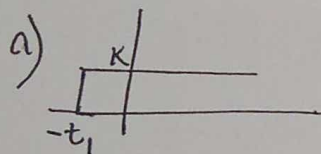
$f(\infty) = \lim_{s \rightarrow 0} s \cdot F(s) = -\frac{1600}{800} = -2$

Ques



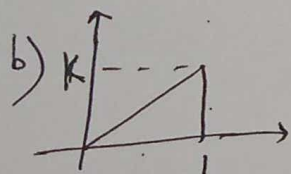
$f(t) = 4u(t) - 2(t-3)u(t-3) + 2(t-5)u(t-5)$

i) Find Lap. of following.



$f(t) = K u(t - (-t_1))$
 $= K u(t + t_1)$

$F(s) = \frac{K e^{t_1 s}}{s}$



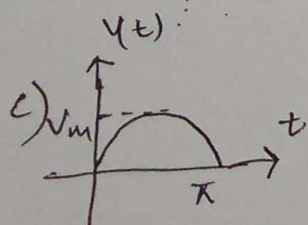
$f(t) = Kt [u(t) - u(t-1)]$

$F(s) = Kt u(t) - K(t-1) u(t-1) - K u(t-1)$
 $= \frac{K}{s^2} - \frac{K e^{-1s}}{s^2} - \frac{K e^{-1s}}{s}$

$f(t) = V_m \sin \omega t [u(t) - u(t - \frac{T}{2})]$

$= V_m \sin \omega t u(t) + V_m \sin \omega(t - \frac{T}{2}) u(t - \frac{T}{2})$

$= V_m \left[\frac{\omega}{\omega^2 + s^2} + \frac{e^{-\frac{T}{2}s}}{\omega^2 + s^2} \cdot \omega \right]$



Circuit Analysis by Laplace Tr

1

$$a^n \frac{d^n i}{dt^n} + a_1 \frac{d^{n-1} i}{dt^{n-1}} + \dots + a_{n-1} \frac{di}{dt} + a_n i = v(t)$$

$$\underset{\substack{\uparrow \\ \mathcal{L}[i(t)]}}{I(s)} = \frac{\mathcal{L}[v(t)] + \text{initial condn}}{a_0 s^n + a_1 s^{n-1} + \dots + a_n}$$

$$I(s) = \frac{P(s)}{Q(s)}$$

$$Q(s) = 0 \quad (\text{characteristic eqn})$$

Partial fractions.

(1) If roots of $Q(s)$ are simple

$$I(s) = \frac{K_1}{s+s_1} + \frac{K_2}{s+s_2} + \dots + \frac{K_n}{s+s_n}$$

$$K_j = (s+s_j) \frac{P(s)}{Q(s)} \Big|_{s=-s_j}$$

(2) roots of multiple order.

$$I(s) = \frac{K_{11}}{s+s_1} + \frac{K_{12}}{(s+s_1)^2} + \dots + \frac{K_{1r}}{(s+s_1)^r}$$

$$K_{1r} = (s+s_1)^r \frac{I(s)}{(s+s_1)^r} \Big|_{s=-s_1} \quad K_{11} = \frac{1}{(r-1)!} \frac{d^{r-1}}{ds^{r-1}} \left[(s+s_1)^r I(s) \right] \Big|_{s=-s_1}$$

(3) Complex roots

$$F(s) = \frac{P(s)}{(s+\alpha+j\omega)(s+\alpha-j\omega) Q(s)} = \frac{K_1}{(s+\alpha+j\omega)} + \frac{K_1^*}{(s+\alpha-j\omega)}$$

$$K_1 = (s+\alpha+j\omega) F(s) \Big|_{s=-(\alpha+j\omega)}$$

RG

Ques

Solve

$$x'' + 3x' + 2x = 0, \quad x(0^+) = 2, \quad x'(0^+) = -3.$$

$$s^2 X(s) - s x(0^+) - x'(0^+) + 3s X(s) - 3x(0^+) + 2X(s) = 0$$

$$(s^2 + 3s + 2) X(s) - 2s + 3 - 6 = 0.$$

$$(\cancel{s^2 + 3s + 2}) X(s) = \frac{2s + 3}{s^2 + 3s + 2} = \frac{2s + 3}{(s+1)(s+2)}$$

$$= \frac{K_1}{(s+1)} + \frac{K_2}{(s+2)}$$

$$K_1 = \frac{2s+3}{s+2} \Big|_{s=-1} = \frac{-2+3}{1} = 1$$

$$K_2 = \frac{2s+3}{s+1} \Big|_{s=-2} = \frac{-4+3}{-1} = 1$$

$$X(s) = \frac{1}{s+1} + \frac{1}{s+2}$$

$$x(t) = \mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1}\left[\frac{1}{s+1} + \frac{1}{s+2}\right]$$

$$= e^{-t} + e^{-2t} \quad e^{-at} \cdot \frac{1}{s+a}$$

Transformed Circuit Components Representation.

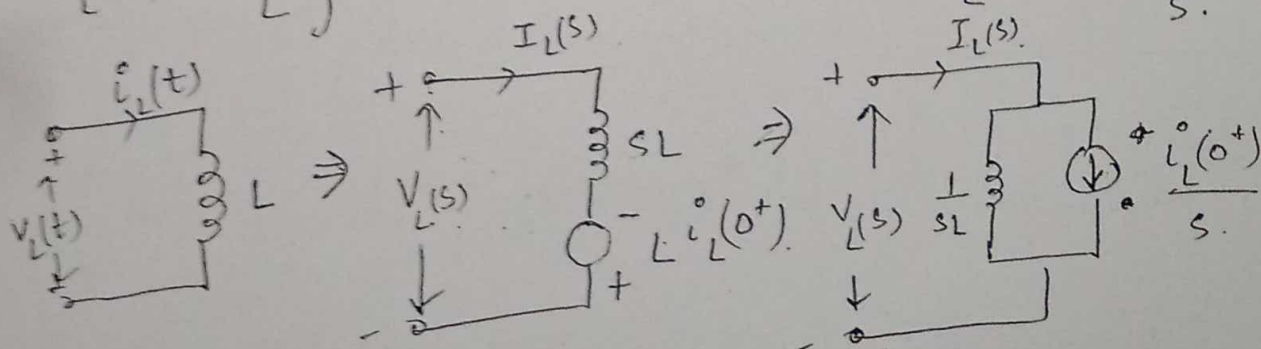
$$\left. \begin{array}{l} v(t) = V(s) \\ i(t) = I(s) \end{array} \right\} *$$

$$R = R$$

Inductor

$$v_L(t) = L \frac{di_L(t)}{dt} \Rightarrow V_L(s) = sL I_L(s) - L i_L(0^+)$$

$$i_L(t) = \frac{1}{L} \int v_L(t) dt + i_L(0^+) \Rightarrow I_L(s) = \frac{1}{sL} V_L(s) + \frac{i_L(0^+)}{s}$$

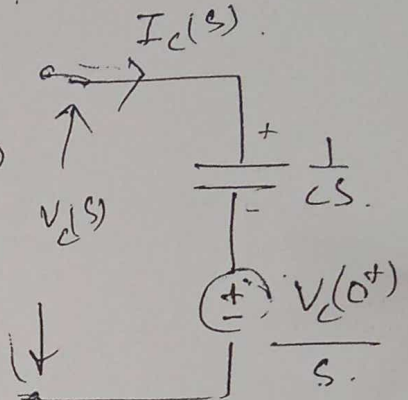
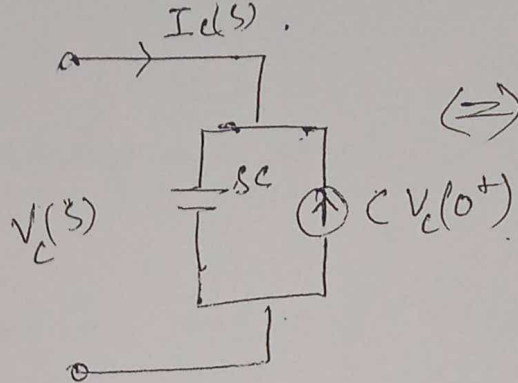
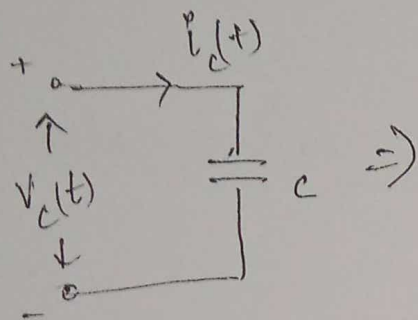


$\int_0^t e^{-as} ds = \frac{1-a}{s}$

Capacitor

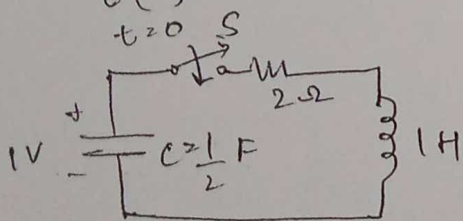
$$i_C(t) = C \frac{dV_C(t)}{dt} \Rightarrow I_C(s) = CS V_C(s) - C V_C(0^+)$$

$$V_C(t) = \frac{1}{C} \int i_C(t) dt \Rightarrow V_C(s) = \frac{1}{CS} I_C(s) + \frac{V_C(0^+)}{s}$$



Consider the R-L-C circuit with $R=4\Omega$, $L=1H$ & $C=1F$ excited by a voltage source $V=1V$. Find the current $i(t)$ & draw s-domain representation of the circuit.

Ques Consider a series R-L-C circuit with the capacitor initially charged to $V=1V$. Find $i(t)$ & draw s-domain representation of the circuit.



$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt + V_C(0^+) = 0$$

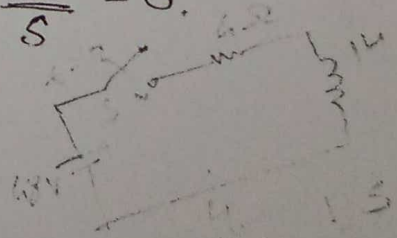
$$L [s I(s) - i(0^+)] + R I(s) + \frac{1}{CS} I(s) + \frac{V_C(0^+)}{s} = 0$$

$$1 [s I(s) - 0] + 2 I(s) + \frac{2}{s} I(s) - \frac{1}{s} = 0$$

$$s I(s) + 2 I(s) + \frac{2}{s} I(s) - \frac{1}{s} = 0$$

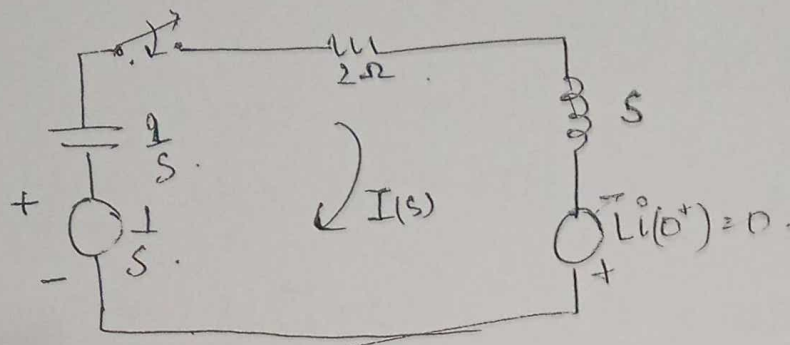
$$I(s) \left(s + 2 + \frac{2}{s} \right) = \frac{1}{s}$$

$$I(s) = \frac{1}{s(s^2 + 2s + 2)} = \frac{1}{s^2 + 2s + 2}$$

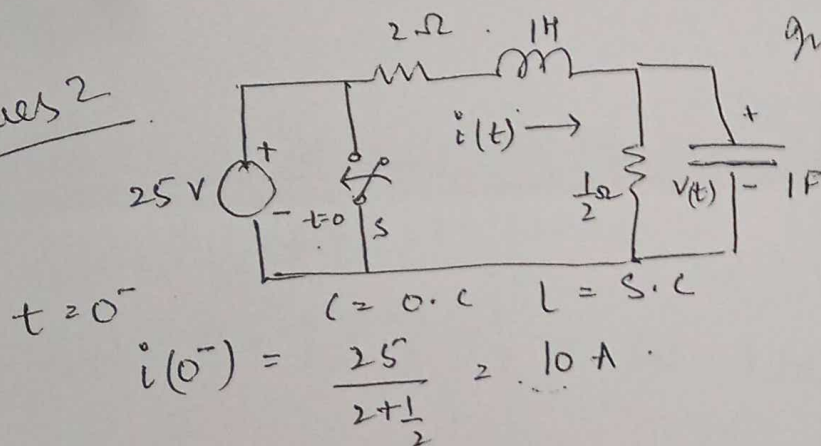


$$I(s) = \frac{1}{(s+1)^2 + 1}$$

$$i(t) = \mathcal{L}^{-1}\{I(s)\} = e^{-t} \sin t$$



Ques 2



In the crkt steady state is reached with S-open. S is closed at $t = 0$. Find $i(t)$ & $v(t)$ for $t > 0$.

At $t = 0$

$$2i + 1 \cdot \frac{di}{dt} + v(t) = 0$$

taking L.T

$$2Is + sI(s) - i(0^+) + v(s) = 0$$

$$(2+s)I(s) - 10 + v(s) = 0$$

$$(2+s)I(s) + v(s) = 10$$

(1)

$$i(t) = \frac{V(t)}{1/2} + \frac{dV(t)}{dt}$$

L.T

$$I(s) = 2V(s) + sV(s) - V(0^+)$$

$$I(s) = (2+s)V(s) - 5$$

$$V(0^+) = V(0^-) = 5 \quad (2+s)I(s) + V(s) = 10 - (1)$$

$$-I(s) + (2+s)V(s) = 5 \quad \text{--- (ii)}$$

Solving (i) & (ii)

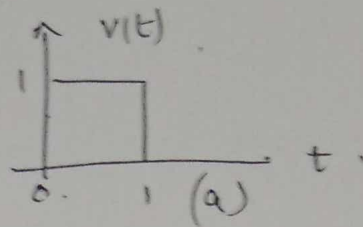
$$I(s) = \frac{10s+15}{(s+2)^2+1} = \frac{10(s+2)}{(s+2)^2+1} - \frac{5}{(s+2)^2+1}$$

$$i(t) = 10e^{-2t} \cos t - 5e^{-2t} \sin t \quad A$$

$$V(s) = \frac{5s+20}{(s+2)^2+1} = \frac{5(s+2)}{(s+2)^2+1} + \frac{10}{(s+2)^2+1}$$

$$V(t) = 5e^{-2t} \cos t + 10e^{-2t} \sin t$$

Q Obtain L.T of



to determine $i(t)$.

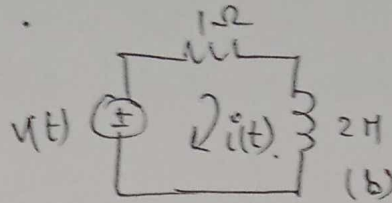
Solⁿ

(a)

$$v(t) = u(t) - u(t-1)$$

$$V(s) = \frac{1}{s} - \frac{1}{s} e^{-s}$$

$$= \frac{1}{s} (1 - e^{-s})$$



$$i(0^+) = 0$$

$$v(t) = 1 \cdot i(t) + 2 \frac{di}{dt}$$

$$V(s) = I(s) + 2s I(s)$$

$$= (1 + 2s) I(s)$$

$$I(s) = \frac{V(s)}{1 + 2s}$$

$$v(t) = u(t-1)$$

$$I_2(s) = \frac{1}{s} e^{-s} \left(\frac{1}{1 + 2s} \right)$$

$$= e^{-s} \left[\frac{1}{s} - \frac{1}{s + \frac{1}{2}} \right]$$

$$i_2(t) = u(t-1) - e^{-\frac{t}{2}} u(t-1)$$

$$= (1 - e^{-\frac{t}{2}}) u(t-1)$$

$$\text{as } v(t) = u(t)$$

$$I_1(s) = \frac{1}{s} \cdot \frac{1}{1 + 2s}$$

$$= \frac{1}{s} - \frac{1}{s + \frac{1}{2}}$$

$$i_1(t) = (1 - e^{-\frac{t}{2}}) u(t)$$

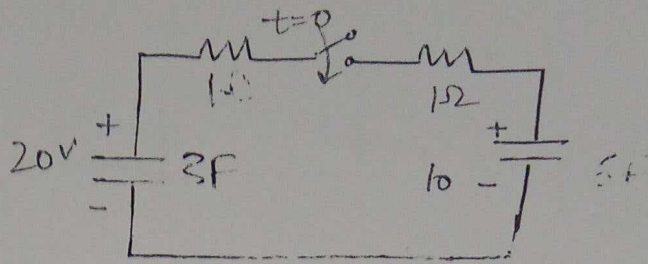
$$= \frac{1}{s} - \frac{e^{-s}}{s(1 + 2s)}$$

$$\therefore i(t) = i_1(t) - i_2(t)$$

$$= (1 - e^{-\frac{t}{2}}) u(t) - (1 - e^{-\frac{t}{2}}) u(t-1) \text{ A}$$

Q Solve for $i(t)$.

& Draw transformed circuit.



Solⁿ

$$\frac{1}{3} \int_{-\infty}^t i(t) dt + 1 i(t) + 1 i(t) + \frac{1}{6} \int_{-\infty}^t i(t) dt = 0$$

$$\frac{1}{3} \int_0^t i dt - 20 + 2 i(t) + \frac{1}{6} \int_0^t i dt + 10 = 0$$

d.T

$$\frac{1}{3s} I_s + 2 I_s + \frac{1}{6s} I_s = \frac{10}{s}$$

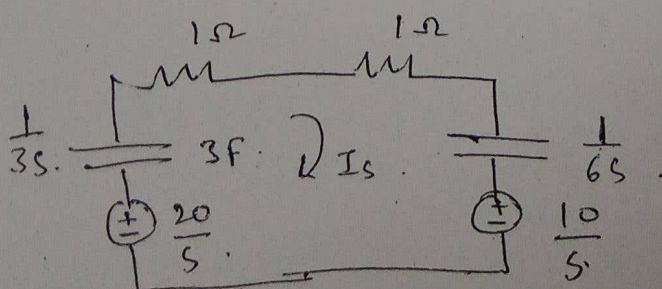
$$\left(\frac{1}{3s} + \frac{1}{6s} \right) + 2 \Big] I_s = \frac{10}{s}$$

$$\left(\frac{2+1+2s}{6s} \right) I_s = \frac{10}{s}$$

$$I_s = \frac{10 \times 6s}{s \times (1+4s)} = \frac{20}{1+4s}$$

$$= \frac{5}{s+0.25}$$

$$i(t) = \mathcal{L}^{-1}[I(s)] = 5 e^{-0.25t} \text{ A.}$$



Transfer functions:

x^{th} funcⁿ of a fixed, linear system is defined as the ratio of d.T of the sys o/p to the d.T of the system input, when all conditions are zero.

$$H(s) = \frac{Y(s)}{X(s)} \Big|_{\text{all initial cond}^n = 0}$$

Ques A sys. has the foll. x^{th} func.

$$\frac{C(s)}{R(s)} = \frac{(s+1)(s+3)}{(s+2)(s^2+8s+32)}$$

Determine time response $x(t)$ of the system for a unit step input.

Solⁿ

$$r(t) = u(t)$$

$$R(s) = \frac{1}{s}$$

$$\text{so } C(s) = \frac{(s+1)(s+3)}{s(s+2)(s^2+8s+32)}$$

$$C(s) = \frac{A}{s} + \frac{B}{(s+2)} + \frac{K(s)+D}{s^2+8s+32}$$

$$\text{effly } A = \frac{1 \times 3}{2(32)} = \frac{3}{64}$$

$$B = \frac{(-2+1)(-2+3)}{-2(4-16+32)} = \frac{-1 \times 1}{-2(20)} = \frac{1}{40}$$

$$(s^2 + 1)(s + 2) = \frac{A}{s+1} (s^2 + 8s + 12) + B(s^2 + 8s + 32) + (Cs + D)(s)(s+2)$$

$$\text{Coeff of } s^3 \Rightarrow 0 = A + B + C$$

$$C = -(A + B)$$

$$= -\left(\frac{3}{64} + \frac{1}{40}\right) = -\frac{23}{320}$$

$$\text{Coeff of } s^2 \Rightarrow$$

$$1 = 8A + 2A + 8B + 2C + D$$

$$1 - (10A + 8B + 2C) = D$$

$$1 - \left(10 \times \frac{3}{64} + \frac{8}{40} + 2 \times -\frac{23}{320}\right) = D$$

$$D = \frac{19}{40}$$

$$C(s) = \frac{3}{64} \cdot \frac{1}{s} + \frac{1}{40} \cdot \frac{1}{s+2} + \frac{-\frac{23}{320} s + \frac{19}{40}}{(s+4)^2 + 4^2}$$

$$= \frac{3}{64} \cdot \frac{1}{s} + \frac{1}{40} \cdot \frac{1}{s+2} + \frac{-\frac{23}{320} \left(s + \frac{19}{40} \times \frac{320}{23} \right)}{(s+4)^2 + 4^2}$$

$$= \frac{-\frac{23}{320} \left(s - \frac{8 \times 19}{23} \right)}{(s+4)^2 + 4^2}$$

$$= \frac{-\frac{23}{320} (s+4) + \frac{23}{320} \left(\frac{8 \times 19}{23} + 4 \right)}{(s+4)^2 + 4^2}$$

$$= \frac{3}{64} \cdot \frac{1}{s} + \frac{1}{40(s+2)} - \frac{23}{320} \frac{s+4}{(s+4)^2 + 4^2} + \frac{61}{320} \frac{4}{(s+4)^2 + 4^2}$$

$$C(t) = \frac{8}{64} + \frac{1}{40} e^{-2t} - \frac{23}{320} e^{-4t} \cos 4t + \frac{61}{320} e^{-4t} \sin 4t. \quad (c)$$